CEBU INSTITUTE OF TECHNOLOGY - UNIVERSITY
College of Computer Studies, IT411 Capstone & Research 2

SMART Goal Traceability & Verification Matrix

Deliverable 04: mapping approved SMART goals to indicators, instruments, and criteria

Course & Section:	IT411 (Section G7) — Capstone & Research 2 | Semester 1, AY 2026–2027
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System Under Evaluation:	playIT Android Application (`v1.2.0-rc2` / `refactor/hear-say-it`, Offline Grade 1 Phonics Tutor)

1. Purpose of This Matrix

In compliance with **Requirement D (**`SMART_Goal_Matrix`**)** of the *Requirements Before Adviser Presentation* (`IT411 | G1 - G8 | SEM1 | 2026-27`), this matrix maps **every measurable outcome** from:

1. **Part A — Originally Approved Proposal & Baseline SRS v2.0 SMART Goals** (`2526-sem2-it332-56_playIT_SRS.pdf` / Project Proposal Revised 4, May 2026), preserving the exact approved wording and measurable targets; and
2. **Part B — Post-MVP Refactored Operational Sub-Objectives (**`HI`**,** `SI`**,** `O3`**–**`O6`**)** (`SRS_v3.0_Refactored.md` & `docs/specs/hear-say-refactor.md`), showing how the deployed system operationalizes each outcome.

> **Methodological Alignment Note (Non-Questionnaire vs. Questionnaire Outcomes):** Per the course directive (*"Remember that not all outcomes should be measured through questionnaires. For example, processing time, task completion rates, error rates, and system performance may require direct observation, system logs, or test results"*), this matrix explicitly assigns **System Telemetry Logs (**`Instrument C`**)**, **Hardware/ADB Benchmarks & Engineering Verification (**`Instrument D`**)**, **Blind Teacher-Rater Acoustic Audits (**`Instrument A` **&** `Instrument B`**)**, and **Silent-Observer Child Session Sheets (**`Instrument B`**)** to all behavioral, accuracy, and latency targets, reserving questionnaires (`Instrument E` TAM/SUS) strictly for stakeholder usability and pedagogical alignment constructs.

2. Part A — Master SMART Goal Matrix (Originally Approved Proposal & Baseline SRS v2.0 Targets)

SMART Goal ID	Originally Approved SMART Goal / Outcome Statement (Verbatim Baseline)	System Module & Requirement ID	Measurable Indicator	Measurement Method (Non-Questionnaire vs. Survey)	Instrument Item / Test Scenario ID	Calculation / Scoring Method	Success Criterion / Target Threshold
SMART-01 *(Specific Obj. 1)*	**Phoneme-to-Grapheme Mapping (26 Letters):** *"Map 26 English alphabet letters into six progressive, frequency-based learning groups following the Marungko-inspired sequence (Group 1: M, S, A, I, O, B through Group 6: J, Q, V, X, Z), unlocking subsequent groups only after completing the preceding group."*	Module 1: Progressive Phonics Map (`FR-01`, `FR-02`, `FR-NEW-01`)	1. Curriculum completeness (26 letters across 6 groups + 6 Blend It milestones = 32 nodes). 2. Group-gate enforcement accuracy (100% sequential lock/unlock integrity). 3. Teacher curriculum alignment rating.	**Non-Questionnaire:** Automated Unit/Integration Test (`GroupUnlockManagerTest`, `UnlockManagerTest`) + Teacher Content Audit (`Instrument A`). **Survey:** Teacher Pedagogical Alignment (`Instrument A` Part 3).	`Instrument A` Items `A1-01..A1-26`, `A2-01..A2-33`, `A3-PA1..PA2`; `Instrument D` Scenario `D-06` (Curriculum & Gate Integrity).	Gate Pass Rate = Verified Transitions / 32 Nodes × 100% Teacher Likert Mean barx = 1/N Σ x_i	**100% (26/26 letters & 6/6 group locks verified)**; **100% (26/26)** phoneme clips pass Gate 3 teacher audit; Teacher Alignment Mean ≥ 4.00 / 5.00.
SMART-02 *(Specific Obj. 2 — Hear It)*	**Multisensory Phoneme Modeling (Hear It):** *"Provide multisensory instruction for each letter through clear phoneme audio pronunciation, synchronized mouth-shape articulation guidance, and contextual object association within ≤ 5.0 seconds per clip at 100% playback accuracy."*	Module 2: Hear It Stage (`FR-03`, `NFR-Perf-02`, `NFR-AUD-01`)	1. Audio clip duration (≤ 5.0 s) & zero schwa intrusion (/■/). 2. Audio-visual synchronization & playback reliability (100%). 3. Child independent completion of 3-step sequence (`Sound` → `Word` → `Prompt`).	**Non-Questionnaire:** Acoustic file inspection (`AudioCompletenessCheckTest`) + Blind Teacher Phoneme Audit (`Instrument A`) + Child Silent-Observer Log (`Instrument B`).	`Instrument A` Part 1 (`26` blind phoneme clips); `Instrument B` Section B1 (`HearIt` step completion); `Instrument D` Item `D-03` (Clip length ≤ 5.0 s).	Phoneme Audit Pass = Clips Rated Pure / 26 × 100% max(Duration$_i$) ≤ 5000 ms	**100% (26/26)** clips ≤ 5.0ext{ s} and rated schwa-free by ≥ 2/3 teachers; **100%** playback completion without crash; ≥ 90% child completion of Hear It without adult help.
SMART-03 *(Specific Obj. 2 — Say It)*	**Offline Speech Recognition & Articulation Verification (Say It):** *"Evaluate the learner's spoken pronunciation offline using the embedded Vosk speech recognition engine with ≥ 75% baseline recognition accuracy (tightened post-MVP to ≥ 80% teacher-ASR agreement and ≤ 15% false-reject rate) and response latency ≤ 0.5 seconds (500 ms) across up to 3 guided attempts."*	Module 2: Say It Stage & `SpeechValidator` (`FR-04`, `NFR-Perf-01`, `NFR-ACC-01`)	1. End-of-speech to UI feedback latency (L_(SayIt) in ms). 2. Teacher-ASR binary agreement rate (P_(agree)). 3. False-reject rate on teacher-certified correct child utterances (P_(FR)). 4. Foil rejection rate (letter names `"em"` & vowel-added `"muh"`).	**Non-Questionnaire:** System Telemetry timestamps (`Instrument C SayItAttempt`) + Blind Dual-Teacher Rating of live child utterances (`Instrument B` Part B2) + Hardware Latency Benchmark (`Instrument D D-01`).	`Instrument B` Part B2 (Trial-by-Trial Say It Rating Log); `Instrument C` (`SayItAttempt` table & latency log); `Instrument D` Item `D-01` (≤ 500 ms) & `D-10` (≤ 100 ms mic cue).	P_(agree) = (TP + TN) / Total Trials P_(FR) = FN / (TP + FN) L_(SayIt) = P90(t_(feedback) − t_(EOS))	**P_(agree) ≥ 80.0%** (95% Wilson CI reported); **P_(FR) ≤ 15.0%**; **100%** rejection of letter-name/schwa foils in unit & spike suite; **P90 Latency ≤ 500ext{ ms}**.

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SMART-04 *(Specific Obj. 2 — Find It & Blend It)*	**Visual-Auditory Letter Discrimination & Word Blending (Find It & Blend It):** *"Reinforce grapheme-phoneme association through a 3-round interactive discrimination game (Find It) with ≤ 0.3 seconds (300 ms) tap-to-feedback response and a decodable CVC word-blending stage (Blend It) at the end of each group."*	Module 2: Find It (`FR-05`, `NFR-Perf-03`) & Module 2b: Blend It (`FR-NEW-01`, `FR-13`)	1. Find It tap-to-feedback latency (L_(FindIt) in ms). 2. Find It round completion rate & first-tap accuracy. 3. Blend It CVC word completion rate across 3 words per group.	**Non-Questionnaire:** Room Database logs (`Instrument C FindItAttempt`, `BlendItAttempt`) + Child Observation Sheet (`Instrument B`) + Touch Latency Test (`Instrument D D-02`).	`Instrument B` Section B1 (`FindIt` & `BlendIt` observation); `Instrument C` (`FindItAttempt` & `BlendItAttempt` records); `Instrument D` Item `D-02` (≤ 300 ms).	L_(FindIt) = P90(t_(audio/visual) − t_(tap)) Accuracy = Correct First Taps / Total Rounds × 100%	**P90 Tap Latency ≤ 300ext{ ms}**; **≥ 80.0%** of Grade 1 learners complete Find It & Blend It rounds with ≥ 2star mastery.
SMART-05 *(Specific Obj. 3 & 4 — Gamification)*	**Gamified Scaffolding, Hearts, Stars & Streaks:** *"Maintain learner engagement through a 5-heart session pool (deducting hearts only on Find It / Blend It errors, never on Say It speech attempts, with a 3-heart recovery on lesson restart), 1-to-3 star mastery ratings (0 lost = 3star, 1–2 lost = 2star, ≥ 3 lost = 1star), and 5/10/15/20-day streak badges."*	Module 3: Heart System (`FR-06`, `FR-07`, `FR-08`) & Module 4: Star & Streak Rewards (`FR-09`)	1. Deterministic accuracy of heart deductions, 3-heart refill, star calculation, and streak increments (100%). 2. Zero punitive heart loss during Say It. 3. Child behavioral engagement & voluntary replay rate.	**Non-Questionnaire:** Domain Unit Suite (`HeartManagerTest`, `StarCalculatorTest`, `StreakTrackerTest`) + Child Behavioral Observation & Exit Choice (`Instrument B` & `Instrument E` Part E1).	`Instrument B` Section B1 (Affective & Heart-Refill Observation); `Instrument E` Items `E1-01`..`E1-03` (Smileyometer & Voluntary Replay Prompt).	Rule Compliance = Passed State Transitions / Total Tested × 100% Replay Rate = Learners Choosing Another Letter / N_(children)	**100% mathematical & state compliance** across all heart/star/streak rules; **0%** Say It heart deductions; **≥ 80.0%** positive child affect / voluntary replay intent.
SMART-06 *(Specific Obj. 5 — Profiles & Offline Persistence)*	**Multi-Profile Management, Arithmetic Gate & 100% Offline Persistence:** *"Support up to 6 local learner profiles on a single shared device with avatar-first child onboarding, a randomized single-digit arithmetic gate protecting the Parent Zone, and 100% offline SQLite/Room data persistence across app restarts without internet connectivity."*	Module 5: Profile & Session Management (`FR-10`, `FR-11`) & Offline Persistence (`NFR-Off-01`, `NFR-Off-02`, `NFR-Sec-01`..`03`)	1. Airplane-mode module operability (4/4 modules: Hear It, Say It, Find It, Blend It + PDF Export). 2. Force-close persistence reliability (20/20 cycles with 0 data loss). 3. Parent gate child-blocking vs. adult-entry success rate.	**Non-Questionnaire:** 20-Cycle Force-Close Persistence Test in Airplane Mode (`Instrument D D-05`, `D-06`) + AndroidManifest Static Privacy Check (`ManifestPrivacyTest`).	`Instrument D` Items `D-05` (20× Force-Close SQLite check), `D-06` (Airplane Mode 4/4 modules), `D-11` (Zero network permissions).	Persistence Rate = Intact DB Cycles / 20 × 100% Offline Operability = Working Modules / 4 × 100%	**100% (20/20)** force-close cycles retain exact stars, hearts, and streaks; **100% (4/4)** modules function in Airplane Mode; **0** network permissions in `AndroidManifest.xml`.

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SMART-07 *(Specific Obj. 6 — Parent Zone & PDF Report)*	**Diagnostic Progress Analytics & Offline PDF Report Generation:** *"Provide parents and teachers with an offline dashboard categorizing each phoneme as Mastered (*`Green` *≥ 80%), Developing (*`Yellow` *60–79%), or At-Risk (*`Red/Amber` *< 60% or > 3 failed attempts), 7-day retention tracking, and native A4 PDF progress report export."*	Module 6: Progress Tracking & PDF Reporting (`FR-12`, `FR-13`)	1. Classification parity between Room DB attempts, `ParentDashboardScreen`, and exported PDF (`100\%`). 2. Caregiver task success rate in identifying a learner's weak phonemes and exporting the PDF. 3. Stakeholder SUS & TAM usefulness scores.	**Non-Questionnaire:** Automated Calculation Tests (`LetterStatusCalculatorTest`, `RetentionCalculatorTest`, `ParentDashboardViewModelTest`) + Caregiver Task Walkthrough. **Survey:** Stakeholder SUS & TAM (`Instrument E` Part E2).	`Instrument D` Item `D-12` (PDF generation & palette parity); `Instrument E` Part E2 (`SUS-01..10`, `TAM-PU1..PU4`, `TAM-PEOU1..PEOU4`).	SUS = 2.5 × Σ Adjusted Item Scores in [0, 100] TAM Mean = 1/k Σ x_i in [1, 5]	**100%** dashboard-to-PDF metric parity; **100%** of parents/teachers successfully export & interpret PDF report; **Mean SUS ≥ 75.0** ("Good" to "Excellent"); **Mean TAM ≥ 4.00 / 5.00**.

3. Part B — Detailed Operational Sub-Objectives Matrix (Post-MVP Refactored Requirements `HI`, `SI`, `O3`–`O6`)

To ensure complete traceability between the **Weeks 1–2 MVP Validation Findings** and the **Week 12 Final User Validation**, the table below maps every granular operational sub-objective from `docs/specs/hear-say-refactor.md` and `docs/SRS_v3.0_Refactored.md`:

Sub-Objective ID	Operational Requirement & Refactored Target	Primary Evidence Type	Instrument & Item / Scenario Reference	Exact Formula / Statistical Estimator	Pass / Acceptance Criterion
HI-1 *(O1a)*	**Phonetic Purity of Isolated Letter Sounds:** All 26 isolated phoneme clips (`audio/phonemes/{letter}.mp3`) contain pure target phonemes with zero intrusive schwa (/■/) or letter-name confusion.	Blind Expert Acoustic Audit (Non-Survey)	`Instrument A` — Part A1 (26-Clip Blind Teacher Transcription & Clarity Sheet)	$Pass_i$ = mathbbI(Σ_(r = 1)^R Correct_(i,r) ≥ lceil 2R/3 rceil)	**26 / 26 clips (100%)** pass blind transcription and receive ≥ 2/3 teacher approval (Gate 3).
HI-2 *(O1b)*	**Comprehension of Hear It Modeling Sequence:** Grade 1 learners complete the 3-step `Hear It` sequence (`Sound` → `Example Word` → `Lily Prompt`) and tap `Next` without adult intervention.	Direct Behavioral Observation + System Log	`Instrument B` — Part B1 (`HearIt` Prompt Count) & `Instrument D` (`D-03`)	P_(HI-2) = (Sessions with Prompts_(HearIt) = 0) / N_(sessions) × 100%	**≥ 90.0%** of observed child lessons complete `Hear It` with 0 adult prompts.
SI-1 *(O2a)*	**Guided Echo & Independent Recall Completion:** Learners produce a valid target word/sound in `Say It` within the 3-attempt scaffolded window (Attempt 1 prompt → Attempt 2 slow model + mouth cue → Attempt 3 model-and-move-on).	System Telemetry + Child Observation	`Instrument C` (`SayItAttempt` table) & `Instrument B` Part B2	P_(SI-1) = (Lessons Cleared in ≤ 3 Attempts) / Total Lessons × 100%	**≥ 80.0%** of lessons achieve accepted production within Attempts 1–3; **100%** advance cleanly after Attempt 3 without deadlock.
SI-2 *(O2b)*	**Articulatory Scaffolding Effectiveness:** When a learner misses Attempt 1, the synchronized mouth-shape cue (`ArticulationGroup`) and slow model on Attempt 2/3 convert initial misses into correct productions.	System Telemetry + Observation	`Instrument B` Part B2 (Attempt 1 Miss → Attempt 2/3 Recovery) & `Instrument D` (`D-09`)	P_(recovery) = Correct on Att. 2 or 3 mid Miss on Att. 1 / Total Misses on Att. 1 × 100%	**100%** of 7 `ArticulationGroup` mouth cues render on Attempt ≥ 2; positive recovery rate documented with 95% Wilson CI.

Sub-Objective ID	Operational Requirement & Refactored Target	Primary Evidence Type	Instrument & Item / Scenario Reference	Exact Formula / Statistical Estimator	Pass / Acceptance Criterion
SI-3 *(O2c)*	**Vosk ASR Judge Accuracy vs. Human Teachers:** Offline `SpeechValidator` agrees with blind human teacher ratings at ≥ 80% and keeps false rejections of correct child speech at ≤ 15%.	Dual-Rater Blind Audit vs. ASR Log	`Instrument B` Part B2 (Teacher Rating T in {0,1} vs. App Decision A in {0,1})	Agreement = (TP + TN) / N_(trials), FRR = FN / (TP + FN), Cohen's kappa	**Agreement ≥ 80.0%**; **False-Reject Rate ≤ 15.0%**; Inter-rater kappa reported.
SI-4 *(O2d)*	**Non-Punitive Affective Safety in Speech Practice:** `Say It` never deducts hearts (`hearts = null`), never flashes harsh red (`#D63031`), and uses warm growth-mindset guidance (`"Good try! Watch my lips..."`).	Static Code/UI Audit + Child Affective Observation	`Instrument B` Part B1 (Hesitation/Refusal tally) & `Instrument D` (`D-08`)	Heart Loss_(SayIt) = 0; Refusal Rate = Mic Refusals / N_(children)	**0 hearts lost** in `Say It`; **0 harsh red** tokens in child UI; **≤ 10.0%** mic refusal rate.
O3 *(Ergonomics)*	**Pediatric Touch Ergonomics & Independent Navigation:** All interactive child controls meet the ≥ 64 dp × 64 dp touch target floor and ≥ 16 sp (map) / ≥ 24 sp (lesson) typography floor across 4 screen classes.	Automated Layout Matrix Test + Silent-Observer Log	`Instrument D` (`D-07`, `D-08` via `LayoutMatrixTest`) & `Instrument B` Part B1	min(Width, Height) ≥ 64 dp; P_(indep) = Zero-Prompt Screen Transitions / Total Transitions	**100%** of child controls pass ≥ 64 dp bounds check across 4 viewports; **≥ 85.0%** independent child screen transitions.
O4 *(Performance)*	**Budget-Hardware Responsiveness (`SM-A217F`):** Cold start ≤ 2500 ms, Say It latency ≤ 500 ms, Find It latency ≤ 300 ms, frame jank ≤ 10% at 60 FPS, and peak PSS memory ≤ 200 MB on a 3 GB RAM device.	Hardware ADB Benchmark & Telemetry	`Instrument D` Items `D-01`, `D-02`, `D-04`, `D-10` (`adb shell dumpsys gfxinfo` / `meminfo`)	P50 / P90 / Max latency (ms); Jank % = Janky Frames / Total Frames × 100%	**Say It P90 ≤ 500ext{ ms}**; **Find It P90 ≤ 300ext{ ms}**; **Jank ≤ 10.0%**; **0 ANRs / 0 crashes** over 5,000 monkey events.
O5 *(Offline & Privacy)*	**Zero-Network Sovereignty & Child Data Privacy:** Complete offline execution in Airplane Mode with zero network permissions and strict adherence to RA 10173 (Data Privacy Act of 2012).	Manifest Inspection + Airplane Mode Stress Test	`Instrument D` Items `D-05`, `D-06`, `D-11` (`ManifestPrivacyTest`)	Network Permissions = 0; Cloud Calls = 0	**0** `INTERNET` **permissions**; **20/20** force-close cycles pass; **100%** local-only storage.
O6 *(Stakeholder Value)*	**Perceived Pedagogical Usefulness, Usability & Diagnostic Utility:** Teachers and parents validate that playIT's Marungko sequence, Blend It decoding, and PDF reports deliver actionable early-literacy value.	Standardized Questionnaires (SUS + TAM) + Task Completion	`Instrument E` Part E2 (`SUS-01..10`, `TAM-01..10`) & `Instrument A` Part A3	Brooke SUS (0–100); 5-Point Likert Mean & SD per construct	**SUS ≥ 75.0**; **TAM Perceived Usefulness ≥ 4.00 / 5.00**; **TAM Ease of Use ≥ 4.00 / 5.00**.